Simulation D | Common Resources

Let's play a little game. So far in this class we've focused on Private Goods: goods that are rival (meaning my using them makes less available for you) but also excludable (meaning I can keep my goods away from you).

We talked about externalities, which were goods with some non-excludability: either 1) you couldn't stop me from benefiting from your good, or 2) I couldn't stop you from benefiting from mine. This was the externality.

Today we're going to take this idea to the limit, by looking at goods which have no private component, which carry only externalities: Common Resources.

Here's the game. You operate a small scale fishing operation with a few of your friends. The lake you fish on is big but has a finite space with a finite population of fish: Ramora, which is a magical fish native to the Indian Ocean but also is able to survive some fresh water lakes. Some facts about fishing the lake:

1. Every month the fish spawn, doubling in population.
2. Every month you operate your fishing operation, you have to pay for equipment maintenance: your monthly costs, equal to the revenue from 1000 fish. So to continue to operate, you need to catch at least one.
3. There isn't much information about the number of fish in the lake.
4. The carrying capacity of the lake for fish is 20000 fish.

The lake is this jar. In the jar are pennies, each representing 1000 fish.

Mention Elinor Ostrom at some point.

Do three rounds:

1. No communication, no visibility
2. Communication, no visibility
3. Communication, visibility

Notes

I used groups of 5, which was too small to demonstrate tragedy of the commons for many classes.